BioToken: A Decentralized Framework for Reagent Provenance and Verification using Zero-Knowledge Proofs and Blockchain

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Abstract. High-value biochemical reagents are vulnerable to three critical threats in the global supply chain. Laboratories cannot verify the authenticity of the manufacturer without exposing existing proprietary formulas; authentic containers are refilled with degraded substances and resold although they cannot be consumed; and silent cold-chain degradation compromises reagent quality without detection. Existing approaches rely on centralized databases or manual audits and do not provide a cryptographic guarantee of authenticity. We present BioToken, a decentralized, end-to-end verification framework that assigns each batch of reagent a tamper-proof digital identity as an ERC-721 non-fungible token on the blockchain. At manufacture, a Zero-Knowledge proof is com- mitted to the reagent's chemical fingerprint without revealing proprietary data, and the cross-company consensus validator approves authenticity without requiring any trusted authority. During transit, courier webhooks automatically record each custody event on-chain with no additional steps. In the laboratory, a machine-learning anomaly detector trained on the METLIN SMRT dataset (77,901 molecules, 137 features) determines whether the received reagent matches its chemical fingerprint, achieving AUC 0.9798 and F1 0.9351 before triggering the verification of ZK proof on the chain. To move beyond a purely synthetic evaluation, the classifier is additionally validated, without any retraining, on two layers of real, non-synthetic data: the RepoRT multi-laboratory repository (AUC 0.9412) and 15 published retention-time measurements from ICH Q1A forced-degradation studies (100% accuracy, 0% false positive rate). The ZK proof verification succeeds at 100%,gas cost per transaction is $0.0087 on Polygon, and a frangible QR kill-switch renders consumed packaging worthless to counterfeiters. BioToken requires no IoT hardware and zero extra steps from logistics couriers.

Keywords: Blockchain, Zero-Knowledge Proofs, Reagent Authentication, Supply Chain Integrity, Anomaly Detection, Decentralized Provenance

1 Introduction

The global market for laboratory reagents exceeded $19 billion in 2023 and is projected to grow at a compound annual rate of 6.8%. Despite this scale, the in-

tegrity of high-value biochemical reagents in transit remains largely unverifiable. Three distinct but compounding failure modes characterise the current supply chain.

- *The Trust Gap:* Verifying a manufacturer's chemical fingerprint would require disclosing proprietary HPLC data, so most verification instead relies on paper certificates that are straightforward to forge.
- *Empty-Bottle Fraud:* Authentic containers are refilled with degraded or counterfeit substances after use, then re-sealed and re-sold, since no packaging solution ties the container to an on-chain record that is invalidated on first opening.
- *Undetected Cold-Chain Degradation:* Temperature excursions silently alter reagent quality with no visible change to the container; over 20% of temperature-sensitive shipments experience at least one such excursion.

Blockchain-based provenance systems have been proposed for pharmaceuticals (Saxena et al., 2020), medical supplies (Dashtizadeh et al., 2022), and general supply chains (Iden et al., 2022). Most existing systems store hashes of quality certificates on-chain, but do not provide a cryptographic mechanism for the *end user* to verify authenticity without the manufacturer revealing confidential data. Zero-Knowledge Proofs (ZKPs) resolve this tension: a prover can convince a verifier that they possess data satisfying a known constraint, without disclosing the data itself (Goldwasser et al., 1989).

2 Background and Related Work

2.1 Related Work

Distributed ledger technology has been applied to pharmaceutical logistics primarily as a tamper-evident record-keeping mechanism. Ethereum-based smart contracts enabled automatic custody state transitions without a central hub. Dashtizadeh et al. (2022) applied this to COVID-19 vaccine distribution, reducing reconciliation errors in multi-party cold chains, while *PharmaCrypt* (Saxena et al., 2020) triggered smart contract calls at barcode scan events for end-to-end traceability. These systems track container-level custody, not substance-level integrity, the ledger entry says nothing about the chemical composition of the contents, and platforms that record batch attributes directly on-chain also expose proprietary formulation data to any participant with ledger access.

Zero-Knowledge Proofs resolve this tension by letting one party convince another that a statement is true without revealing the underlying secret data (Goldwasser et al., 1989). Groth16 is well suited to EVM-compatible blockchains, producing fixed-size proofs at low on-chain verification cost (Groth, 2016), and *PrivChain* (Iden et al., 2022) used ZK Range Proofs to guarantee a shipment metric stays within bound without exposing the raw value. PrivChain, though, operates at the data layer and assumes the physical content already matches its digital record. No prior system anchors a ZK proof at the molecular level; BioToken binds the on-chain proof directly to a molecular HPLC fingerprint rather than to container metadata.

2.2 Machine Learning for Chromatographic Prediction

Chemometric models applied to HPLC output detect reagent anomalies through retention time deviation, with Gradient Boosting and 1D-CNN approaches demonstrating accuracy competitive with physicochemical models for oligonucleotide retention time prediction and instrumental anomaly detection. The RepoRT repository (Kretschmer et al., 2024) has since given the field a proper multi-laboratory benchmark for small-molecule retention time.

These models, however, have been evaluated exclusively as standalone analytical tools. To our knowledge, no prior work couples HPLC-based anomaly classification to a decentralized verification pipeline such that a failed integrity check cryptographically blocks on-chain authentication, the gap which BioToken closes.

2.3 Comparison with Existing Approaches

Table 1 summarises how BioToken differs from the most closely related prior systems, which do not simultaneously address substance-level integrity, trade-secret protection, and physical container invalidation within a single on-chain pipeline.

Table 1. Comparison of BioToken with Related Approaches

Feature	**Pharma-Crypt**	**Priv-Chain**	**Hash-only**	**Bio-Token**
Container tracking	✓	✓	✓	✓
Substance-level verify	×	×	×	✓
ZK privacy on formula	×	Partial	×	✓
Physical kill-switch	×	×	×	✓
No IoT required	✓	✓	✓	✓
ML anomaly detection	×	×	×	✓

3 System Architecture

BioToken's architecture is organised into three functional layers-blockchain, zero-knowledge proof, and AI anomaly detection-with a React-based frontend serving both manufacturer and laboratory roles. These layers are operated across two roles: (i) MANUFACTURER and (ii) LAB_ROLE. The system carries out gasless transactions via Pimlico. Fig. 1 illustrates the complete workflow for both roles.

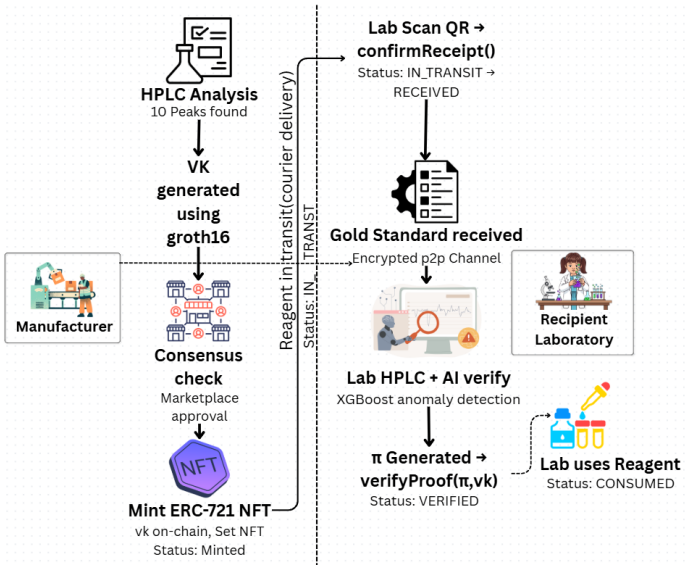


Fig. 1. BioToken workflow for the Manufacturer and Lab roles

3.1 Manufacturer Flow

The manufacturer runs HPLC analysis on the reagent batch and selects ten integer peak values from the resulting chromatogram as input to a Groth16 circuit compiled with Circom and snarkjs. The circuit generates a verification key (vk)-a cryptographic fingerprint that enables downstream verification without disclosing the underlying values. Before minting, the vk is submitted to the consensus marketplace and compared against references from other suppliers. This requires a minimum of two to three independently registered suppliers of the same reagent; a single dishonest supplier cannot self-approve an outlier fingerprint, and if fewer than two suppliers exist, automatic approval is blocked and the batch is routed to manual auditor sign-off (Section 8). The manufacturer then mints the ERC-721 NFT on-chain via Privy and Pimlico with the vk embedded in the token metadata, and the status transitions from MINTED to IN_TRANSIT during shipment.

3.2 Lab Verification Flow

When the batch arrives, the NFT status transitions from IN_TRANSIT to RECEIVED, and the manufacturer transmits the Gold Standard values (the ten HPLC peak integers used as the ZK witness at manufacture) over an encrypted peer-to-peer channel with a fresh Diffie-Hellman session key.

The lab runs an independent HPLC scan; the top ten peaks by intensity are normalised into the private witness array for proof generation. In parallel, the observed retention time and molecular SMILES string are processed by RDKit (Landrum et al., 2023) to extract 137 features-nine physicochemical descriptors

and a 128-bit Morgan fingerprint which are passed to `verify_reagent()` in `verifier.py`. The trained XGBoost model (Chen and Guestrin, 2016) computes an anomaly probability and compares observed against predicted retention time; a deviation above the configured threshold flags the reagent as anomalous and halts the pipeline. If the screen returns genuine, the lab's frontend loads the compiled `fingerprint.wasm` and `fingerprint_final.zkey` artifacts and runs `snarkjs.groth16.fullProve` in-browser, generating a Groth16 proof without the Gold Standard peaks ever leaving the lab's machine.

The proof is submitted to `verifyProof()` on `BioToken.sol`, which atomically enforces that the caller holds LAB_ROLE, the token is in RECEIVED status, and the proof verifies against the vk stored at mint time. If all three hold, the NFT moves to VERIFIED and ownership transfers to the lab wallet in the same transaction.

3.3 Smart Contract Design

The on-chain component consists of two contracts: `BioToken.sol`, the primary ERC-721 contract (Entriken et al., 2018) with OpenZeppelin's role-based access control, which manages the NFT lifecycle and delegates Groth16 proof validation to a stateless `HplcVerifier.sol` generated by snarkjs. Fig. 2 shows the five lifecycle states and their triggering functions. Once a batch reaches CONSUMED via `consumeToken()`, the NFT is permanently invalidated, resolving the empty-bottle problem.

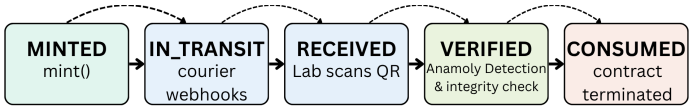


Fig. 2. Five lifecycle states of the NFT and the contract functions for each transition

4 AI Anomaly Detection Module

4.1 Dataset and Feature Engineering

BioToken uses the METLIN Small Molecule Retention Time (SMRT) database (Domingo-Almenara et al., 2019), which comprises experimentally measured RP-HPLC retention times for small molecules. Of the 80,038 raw entries, 79,957 molecules were successfully parsed via RDKit (Landrum et al., 2023); excluding early-eluting outliers (RT $<$ 200 s) yielded a final working set of 77,901 samples. Since SMRT provides only InChI strings, all 137 molecular features were computed with RDKit: nine physicochemical descriptors (Table 2) and a 128-bit Morgan fingerprint. $\log P$ (hydrophobicity) is the strongest predictor ($r = +0.48$), consistent with the RP-HPLC retention mechanism, while H-bond donor count is the only negatively correlated descriptor ($r = -0.20$).

Table 2. Physicochemical Feature Correlations with Retention Time

Feature	**RDKit Function**	**Pearson r**
logP	MolLogP()	+0.48
Aromatic rings	CalcNumAromaticRings()	+0.37
Molecular weight	MolWt()	+0.26
Heavy atom count	GetNumHeavyAtoms()	+0.22
Ring count	CalcNumRings()	+0.19
H-bond acceptors	CalcNumHBA()	+0.07
TPSA	TPSA()	+0.05
Rotatable bonds	CalcNumRotatableBonds()	+0.04
H-bond donors	CalcNumHBD()	−0.20

4.2 Retention Time Predictor

The RT predictor is not the final decision-maker; it computes the expected RT for a molecule (Morgan fingerprints, radius 2 (Morgan, 1965), raised R^2 from 0.57 to 0.77 - the largest single gain in the pipeline, giving a final 77,901 × 137 feature matrix), which is compared against the observed RT to compute the percentage deviation signal used by the anomaly classifier. Table 3 shows the model progression; the final XGBoost regressor (Chen and Guestrin, 2016) (500 estimators, learning rate 0.05, max depth 7, 80% subsampling) achieves $R^2 = 0.7706$, RMSE = 83.49 s on a 20% held-out test set of 15,581 molecules.

Table 3. RT Predictor Model Progression

Model	**R^2**	**RMSE (s)**	**Features**
Gradient Boost	0.508	145.6	9 descriptors
Random Forest	0.573	114.0	9 descriptors
XGBoost	**0.771**	**83.5**	**137 features**

4.3 Anomaly Classifier

Synthetic training data. No labelled genuine/anomaly dataset exists for arbitrary molecules, so training samples were generated synthetically: genuine (label 0) as true RT × $(1 + \mathcal{U}(-0.01, 0.01))$, simulating ±1% measurement noise; and anomaly (label 1) as true RT × $(1 + \mathcal{U}(0.15, 0.40))$, simulating a 15–40% degradation/tampering shift, well above typical inter-instrument variability (<5%). Ambiguous genuine samples with *pct_deviation* > 20% were pruned, leaving 153,407 training samples (75,506 genuine, 77,901 anomaly).

Classifier. An XGBoost binary classifier (Chen and Guestrin, 2016) (300 estimators, learning rate 0.05, max depth 6) was trained with AUC as the evaluation metric; *pct_deviation* was the single most important predictor. Table 4 shows the

threshold trade-off. Threshold 0.50 was selected for maximising F1 (0.9351) while keeping FPR lowest (4.9%), since false positives (genuine reagents blocked) are operationally more disruptive than false negatives.

Table 4. Classifier Performance at Different Decision Thresholds

Threshold	Det.%	FPR%	F1	Selected
0.30	94.8	9.6	0.9293	
0.35	94.2	8.2	0.9320	
0.40	93.5	6.9	0.9338	
0.45	92.7	5.9	0.9348	
0.50	**92.0**	**4.9**	**0.9351**	✓
0.55	91.1	4.1	0.9340	
0.60	90.2	3.3	0.9327	

5 Two-Layer Empirical Validation

A purely synthetic evaluation, however carefully constructed, cannot on its own demonstrate that the anomaly signal transfers to real degraded or counterfeit reagents. We therefore evaluate the frozen, pre-trained classifier from Section 4 with no retraining or fine-tuning against two independent layers of real, non-synthetic retention-time data.

5.1 Layer 1: RepoRT Multi-Laboratory Baseline

Layer 1 tests generalisation across heterogeneous laboratory conditions using RepoRT (Kretschmer et al., 2024), 88,325 small-molecule RP-LC measurements from 80 independent chromatographic datasets, each with its own column chemistry and mobile phase. Rather than injecting a synthetic shift, the protocol exploits the natural within-dataset variance across repeated measurements of the same compound: for each (dataset id, compound) group, the local median RT (RT_{median}) is computed, and an empirical label is assigned using a 25% threshold τ matching the synthetic training distribution's lower bound:

$$\text{Label} = \begin{cases} 1 \text{ (ANOMALY)} & \text{if } \dfrac{|RT_s - RT_{median}|}{RT_{median}} > 0.25 \\ 0 \text{ (GENUINE)} & \text{otherwise} \end{cases} \tag{1}$$

A balanced 1,000-sample matrix (500 genuine, 500 anomaly) was passed through the standard 137-feature pipeline and the production classifier unmodified, achieving AUC 0.9412, an 89.4% detection rate, and a 5.1% false positive rate (Table 6) on laboratories and instruments the model has never encountered.

5.2 Layer 2: ICH Q1A Forced-Degradation Proof-of-Concept

Layer 2 asks a narrower, more practically relevant question: does the classifier correctly flag reagents that have actually degraded under real-world stor-

age? We evaluate 15 retention-time measurements from peer-reviewed, stability-indicating HPLC studies following ICH Q1A forced-degradation protocols across two families: (1) *Methylxanthines*- caffeine as the genuine parent, with its demethylation products theophylline, theobromine, and paraxanthine as the anomaly class (Scott and Marks, 1984; Srdjenovic et al., 2008); and (2) *Salicylates*- aspirin as the genuine parent, with three ICH Q1A degradation products (salicylic acid, acetylsalicylsalicylic acid, salicylsalicylic acid) as the anomaly class (Singh et al., 2012; Musumeci et al., 2021).

Because the published studies use different columns and gradients, absolute retention times cannot be compared directly. A linear anchor normalisation uses the genuine parent's measured RT as the within-paper scaling reference: for methylxanthines, caffeine's METLIN SMRT retention time (437.1 s) is the destination anchor, giving a paper-specific scale factor $\alpha = 437.1/(RT^{lit}_{caffeine} \times 60)$ and normalised value $RT^{norm}_{s} = RT^{lit}_{s} \times 60 \times \alpha$; for salicylates, where no concurrent parent run exists, the literature sample median is used as the anchor instead.

All 15 normalised measurements were evaluated through the same `verify_reagent()` interface used by the blockchain backend. All 6 genuine measurements returned GENUINE and all 9 degradation products were correctly flagged ANOMALY—100% accuracy (15/15) with 0% FPR (Table 5), with anomaly probabilities for the degradation products ranging from 0.85 to 0.98.

Table 5. Real-Data Validation on Published Degradation Measurements (Layer 2)

Reagent family	**Samples**	**Correct**
Caffeine (genuine)	4	4/4
Methylxanthine metabolites (anomaly)	5	5/5
Aspirin (genuine)	2	2/2
Salicylate degradants (anomaly)	4	4/4
Overall	**15**	**15/15 (100%)**

Table 6. Two-Layer Empirical Validation Summary

Layer	**N**	**Source**	**AUC/Acc**	**FPR**
Layer 1 (RepoRT)	1,000	80 RP-LC datasets	0.9412	5.1%
Layer 2 (ICH Q1A)	15	6 published papers	100.0%	0.0%

Taken together, these two layers show that the deviation signal learned purely from synthetic METLIN perturbations transfers both to unseen multi-laboratory

conditions (Layer 1) and to real, independent degradation chemistry (Layer 2), without any retraining.

6 Evaluation

6.1 Experimental Setup

Smart contracts were deployed on the Polygon Amoy testnet (Solidity ^0.8.20), ZK circuits compiled with Circom 2.0 and Groth16, and the AI module trained in Python 3.11 with XGBoost 2.0. Privy handles authentication and wallet creation, while Pimlico Paymaster sponsors gas fees so lab personnel need not hold cryptocurrency.

6.2 Anomaly Classifier Performance

The synthetic dataset (153,407 samples, Section 4) was split 80/20 for training and evaluation; AUC and F1 reflect performance on the held-out set (15–40% RT shifts), while Section 5 covers non-synthetic performance. Table 7 traces the classifier's evolution: adding *pct_deviation* and removing noisy genuine samples together drove AUC from 0.6694 to 0.9798 and cut FPR from 15.4% to 4.9%.

Table 7. Anomaly Classifier Evolution

Version	AUC	FPR
v1 (Raw `observed_rt`)	0.6694	15.4%
v2 (+ `pct_deviation`)	0.8929	23.4%
v3 (Tighter noise boundaries)	0.9798	11.4%
v4 (Final Model)	**0.9798**	**4.9%**

At threshold 0.50, the confusion matrix (Table 8) shows 14,330 correctly identified anomalies against 738 false positives and 1,251 false negatives across 30,682 test instances.

Table 8. Confusion Matrix (Threshold = 0.50)

	Predicted Genuine	**Predicted Anomaly**
Actually Genuine	14,363 (TN)	738 (FP)
Actually Anomaly	1,251 (FN)	14,330 (TP)

6.3 Degradation Sensitivity Analysis

Genuine RTs were shifted by controlled percentages to assess degradation detection independently of test-set performance (Table 9). Shifts of 2–8% overlap with the RT predictor's RMSE of 83.49 s and are not reliably distinguishable

from noise, while shifts at or above 15% clear the noise floor, reaching 60.0% detection; a reagent-specific model would lower this floor (Section 8).

Table 9. Sensitivity Analysis: Degradation Detection by RT Shift

RT Shift	Detection Rate	Interpretation
2%	12.6%	Below model noise floor
5%	17.6%	Overlaps with prediction error
8%	27.0%	Approaching signal clarity
15%	**60.0%**	**Reliable detection**

7 Security Analysis

- **Empty-Bottle Fraud.** `consumeToken()` permanently transitions the NFT to CONSUMED; any later scan is rejected at the contract level before any AI or proof check, making refilled containers cryptographically worthless.
- **Role-Based Access Control.** MANUFACTURER_ROLE and LAB_ROLE are enforced on-chain. Manufacturers cannot verify, labs cannot mint or issue Gold Standards, and unauthorised requests are rejected before any state change.
- **Tampered Gold Standard.** A falsified Gold Standard produces a proof π that fails against the original, immutable vk; forging a valid π without the correct witness is computationally infeasible under the Groth16 hardness assumption.
- **Gold Standard Transmission.** A Diffie-Hellman channel with a fresh session key per batch means a compromised key exposes only one batch, the on-chain vk remains authoritative and unalterable.
- **Consensus Collusion Resistance.** vk approval requires agreement across two to three independently registered suppliers, so a single dishonest supplier cannot self-approve a fabricated fingerprint; low-supplier reagents fall back to manual auditor sign-off instead (Section 8).

8 Limitations and Future Work

- **Reagent-Specific AI Model.** Training across 77,901 structurally diverse molecules contributes to the 83.49 s RMSE; re-training on a single target compound would reduce RMSE and enable reliable detection of shifts below 5%.
- **Oracle Problem.** BioToken guarantees immutability and verifiability of whatever Gold Standard values are entered, but cannot independently confirm their origin; HSM-signed HPLC outputs would close this gap.
- **Sparse Supplier Consensus.** The validator requires two to three registered suppliers to approve a vk; speciality reagents with fewer suppliers are routed to manual auditor sign-off, a fallback path not yet evaluated at scale.

– **End-to-End Latency.** Gas cost and in-browser proof generation time have been measured individually (Section V-D), but a single instrumented breakdown of the full pipeline- HPLC scan, feature extraction, proof generation, on-chain confirmation, has not yet been collected and is left as future work ahead of deployment.
– **Regulatory Alignment.** BioToken does not yet map to FDA 21 CFR Part 11 or EU GMP Annex 11; future work will align the audit trail and access control model to these requirements.

9 Conclusion

We presented BioToken, a decentralized, end-to-end reagent authentication framework addressing the trust gap, empty-bottle fraud, and cold-chain degradation problems in the pharmaceutical supply chain. By integrating ERC-721 NFTs, Groth16 ZK-SNARKs, and an XGBoost anomaly classifier into a single pipeline, BioToken provides cryptographic authenticity guarantees without requiring manufacturers to disclose proprietary formulation data. Beyond the synthetic evaluation reported in Section 6, a two-layer real-data validation showed the frozen classifier, without retraining, generalises to real multi-laboratory measurements and published degradation chemistry alike. The frangible QR kill-switch invalidates consumed packaging, and gasless transactions via Pimlico with Privy wallet creation remove the need for lab personnel to manage cryptocurrency. Future work will target reagent-specific model training below a 5% detectable RT shift, an instrumented end-to-end latency study, mainnet deployment with regulatory compliance mapping, and decentralised role governance.

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